Docker Task

Task Description:

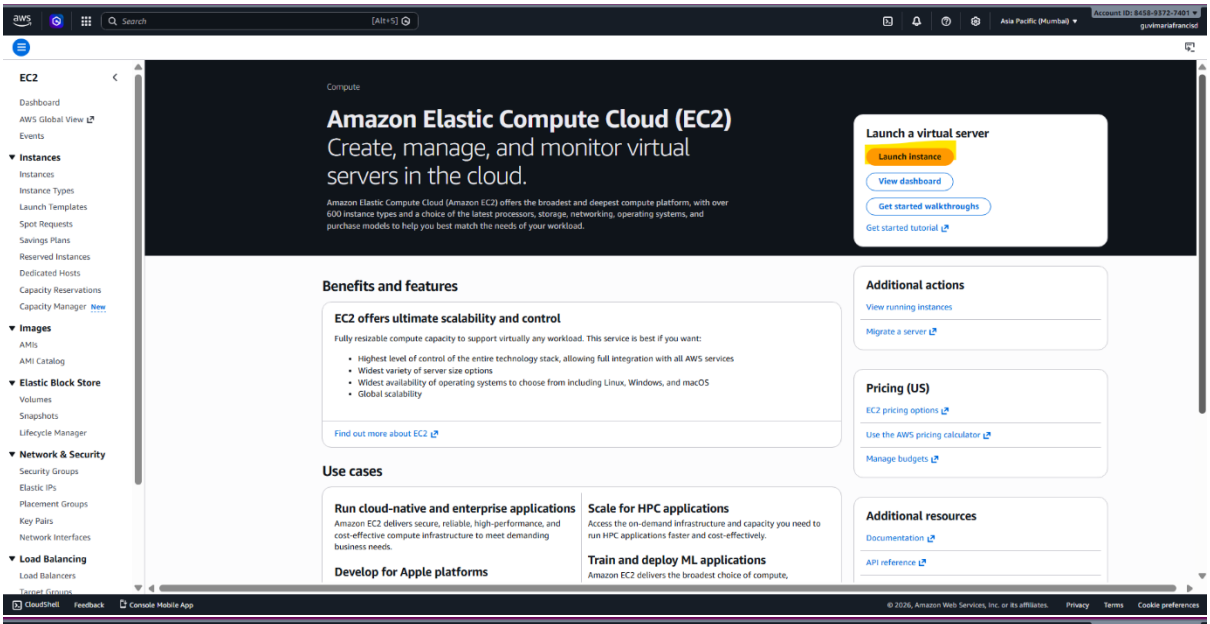
Install docker on EC2 and explore the docker commands

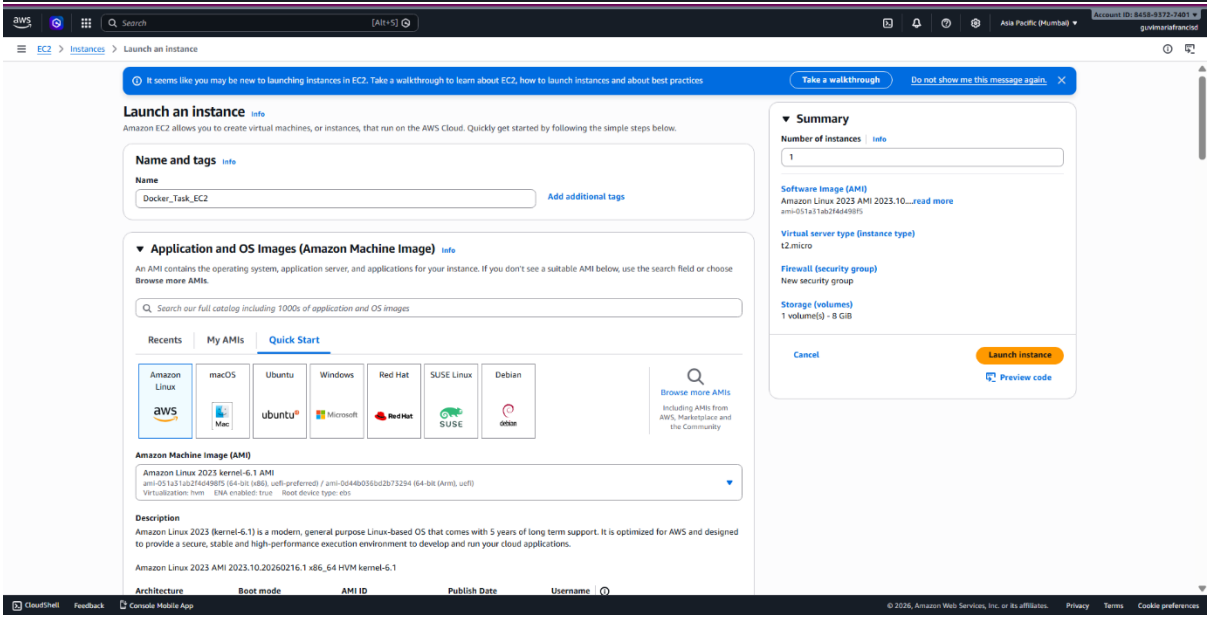
- docker images
- docker containers
- docker volumes
- docker network

Solution:

1. Installing docker on EC2:

Creating an EC2 instance:





EC2 > Instances > Launch an instance

Launching instance
Launch initiation
80%

▸ Details

Please wait while we launch your instance.
Do not close your browser while this is loading.

EC2 > Instances

Instances (1/1) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IP
Docker_Task_EC2	i-08b75b013e8e28e47	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-3-110-151-150.ap-...	3.110.151.150	–	–

i-08b75b013e8e28e47 (Docker_Task_EC2)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID: i-08b75b013e8e28e47
IPv6 address: –
Hostname type: IP name: ip-172-31-12-159.ap-south-1.compute.internal
Answer private resource DNS name: IPv4 (A)
Auto-assigned IP address: 3.110.151.150 [Public IP]
IAM role: –
IMDSv2: Required
Operator

Public IPv4 address: 3.110.151.150 | open address
Instance state: Running
Private IP DNS name (IPv4 only): ip-172-31-12-159.ap-south-1.compute.internal
Instance type: t2.micro
VPC ID: vpc-08ce7316a9289a6fe
Subnet ID: subnet-0a949aba02329e588
Instance ARN: arn:aws:ec2:ap-south-1:845893727401:instance/i-08b75b013e8e28e47

Private IPv4 addresses: 172.31.12.159
Public DNS: ec2-3-110-151-150.ap-south-1.compute.amazonaws.com | open address
Elastic IP addresses: –
AWS Compute Optimizer finding: User: arn:aws:iam::845893727401:user/guvimariafrancisd is not authorized to perform: c ompute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy a llows the compute-optimizer:GetEnrollmentStatus action
Retry
Auto Scaling Group name: –
Managed: false

EC2 > Instances > i-08b75b013e8e28e47 > Connect to instance

Connect Info
Connect to an instance using the browser-based client.

Instance ID: i-08b75b013e8e28e47 (Docker_Task_EC2)
VPC ID: vpc-08ce7316a9289a6fe
Security groups: sg-097b86432c9a080be (launch-wizard-1)
IAM role: –

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

Instance ID
i-08b75b013e8e28e47 (Docker_Task_EC2)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is AmazonLinux.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "AmazonLinux.pem"
4. Connect to your instance using its Public DNS:
   ec2-3-110-151-150.ap-south-1.compute.amazonaws.com

Example:
ssh -i "AmazonLinux.pem" ec2-user@ec2-3-110-151-150.ap-south-1.compute.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Logging into the EC2 instance using SSH:

```
ec2-user@ip-172-31-12-159:~

GHOST@GHOST:~/AWSKey$ chmod 400 "AmazonLinux.pem"
GHOST@GHOST:~/AWSKey$ ssh -i "AmazonLinux.pem" ec2-user@ec2-3-110-151-150.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-110-151-150.ap-south-1.compute.amazonaws.com (3.110.151.150)' can't be established.
ED25519 key fingerprint is SHA256:eulgcdxiFIFxQuSieao9mugTOJUKmT9Sb14ehq65uIM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? Yes
Warning: Permanently added 'ec2-3-110-151-150.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-12-159 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                                    223 kB/s |  31 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
```

After updating the machine, installing docker:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:00:30 ago on Fri Feb 27 10:00:12 2026.
Dependencies resolved.
================================================================================================================================
 Package                                  Architecture            Version                                 Repository          Size
================================================================================================================================
Installing:
 docker                                   x86_64                  25.0.14-1.amzn2023.0.2                  amazonlinux          46 M
Installing dependencies:
 container-selinux                        noarch                  4:2.242.0-1.amzn2023                    amazonlinux          58 k
 containerd                               x86_64                  2.2.1-1.amzn2023.0.1                    amazonlinux          24 M
 iptables-libs                            x86_64                  1.8.8-3.amzn2023.0.2                    amazonlinux         401 k
 iptables-nft                             x86_64                  1.8.8-3.amzn2023.0.2                    amazonlinux         183 k
 libcgroup                                x86_64                  3.0-1.amzn2023.0.1                      amazonlinux          75 k
 libnetfilter_conntrack                   x86_64                  1.0.8-2.amzn2023.0.2                    amazonlinux          58 k
 libnfnetlink                             x86_64                  1.0.1-19.amzn2023.0.2                   amazonlinux          30 k
 libnftnl                                 x86_64                  1.2.2-2.amzn2023.0.2                    amazonlinux          84 k
 pigz                                     x86_64                  2.5-1.amzn2023.0.3                      amazonlinux          83 k
 runc                                     x86_64                  1.3.4-1.amzn2023.0.2                    amazonlinux         3.9 M

Transaction Summary
================================================================================================================================
Install  11 Packages

Total download size: 75 M
Installed size: 282 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm                                   1.4 MB/s |  58 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                                     8.0 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                                      4.4 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                                           2.5 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm                            1.8 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                                     890 kB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                                          3.9 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm                                                2.5 MB/s |  83 kB     00:00
(9/11): runc-1.3.4-1.amzn2023.0.2.x86_64.rpm                                               26 MB/s | 3.9 MB     00:00
(10/11): containerd-2.2.1-1.amzn2023.0.1.x86_64.rpm                                        43 MB/s |  24 MB     00:00
(11/11): docker-25.0.14-1.amzn2023.0.2.x86_64.rpm                                          47 MB/s |  46 MB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                      73 MB/s |  75 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                        1/1
  Installing       : runc-1.3.4-1.amzn2023.0.2.x86_64                                                                      1/11
  Installing       : containerd-2.2.1-1.amzn2023.0.1.x86_64                                                                2/11
  Running scriptlet: containerd-2.2.1-1.amzn2023.0.1.x86_64                                                                2/11
  Installing       : pigz-2.5-1.amzn2023.0.3.x86_64                                                                        3/11
  Installing       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                                                  4/11
  Installing       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                                                             5/11
  Installing       : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                                                    6/11
  Installing       : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                                                             7/11
  Installing       : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                              8/11
  Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                              8/11
  Installing       : libcgroup-3.0-1.amzn2023.0.1.x86_64                                                                   9/11
  Running scriptlet: container-selinux-4:2.242.0-1.amzn2023.noarch                                                         10/11
  Installing       : container-selinux-4:2.242.0-1.amzn2023.noarch                                                         10/11

Installed:
  container-selinux-4:2.242.0-1.amzn2023.noarch          containerd-2.2.1-1.amzn2023.0.1.x86_64        docker-25.0.14-1.amzn2023.0.2.x86_64
  iptables-libs-1.8.8-3.amzn2023.0.2.x86_64              iptables-nft-1.8.8-3.amzn2023.0.2.x86_64      libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64     libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64     libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.3.x86_64                         runc-1.3.4-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-12-159 ~]$
```

Starting the Docker service and checking the status of it: [Running]

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-12-159 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-12-159 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-02-27 10:03:00 UTC; 6s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Process: 28000 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 28001 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 28002 (dockerd)
      Tasks: 7
     Memory: 28.4M
        CPU: 252ms
     CGroup: /system.slice/docker.service
             └─28002 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.408591747Z" level=info msg="Starting up"
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.472967687Z" level=info msg="Loading containers: start."
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.870588354Z" level=info msg="Loading containers: done."
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.890964076Z" level=info msg="Docker daemon" commit=d334795 containerd-snapshotter=false storage-driver=ov
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.891166838Z" level=info msg="Daemon has completed initialization"
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal dockerd[28002]: time="2026-02-27T10:03:00.936065366Z" level=info msg="API listen on /run/docker.sock"
Feb 27 10:03:00 ip-172-31-12-159.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
```

2. Docker Images:

Checking for images in local and pulling nginx from the web. Finally removing the image.

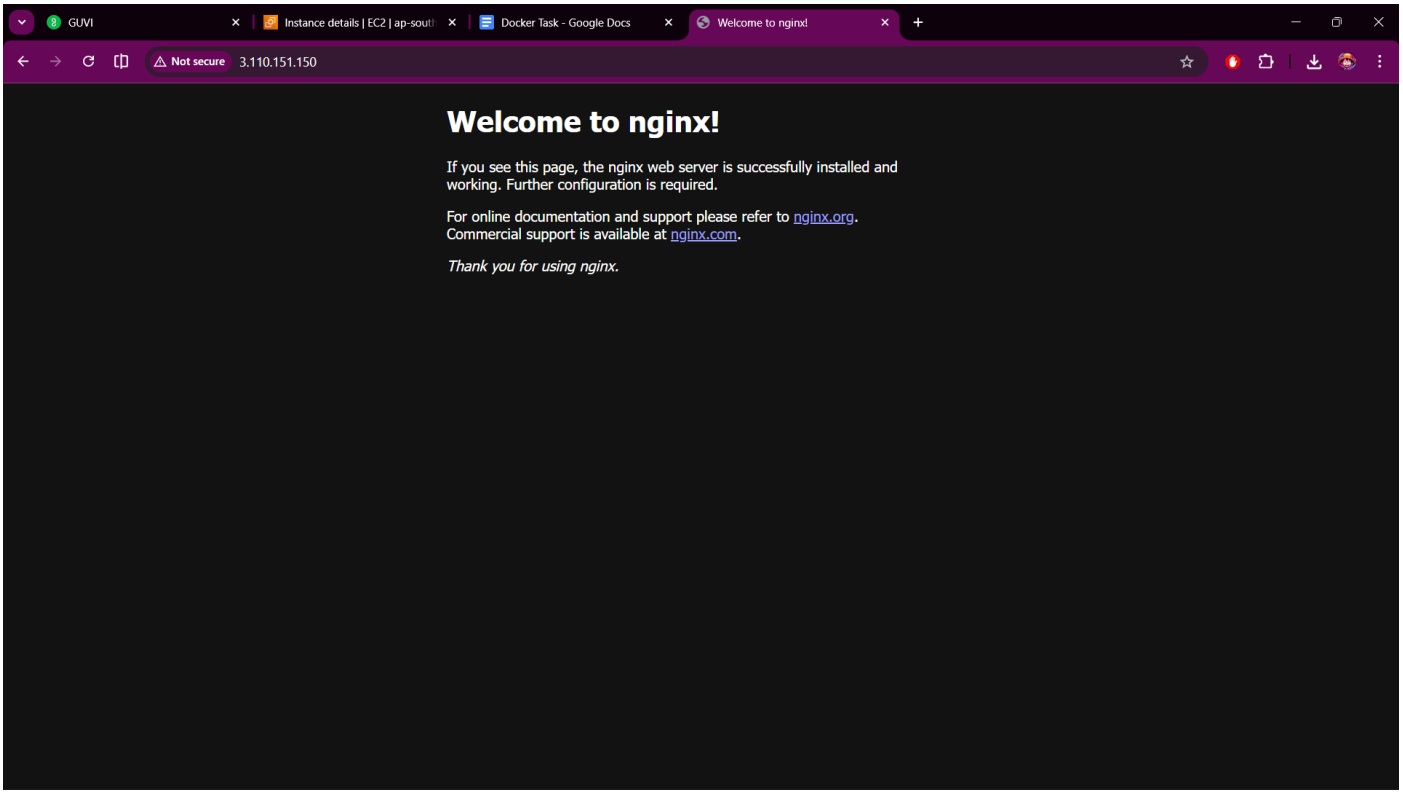
```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
hello-world   latest    1b44b5a3e06a   6 months ago   10.1kB
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker images ls
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
206356c42440: Pull complete
b47f187216b6: Pull complete
1ad233904a11: Pull complete
eedda9fd8786: Pull complete
35ff83c394d6: Pull complete
17d0911eaf62: Pull complete
df0b66c867e4: Pull complete
Digest: sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-12-159 ~]$ docker images ls
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
[ec2-user@ip-172-31-12-159 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    fd204fe2f750   2 days ago     161MB
hello-world   latest    1b44b5a3e06a   6 months ago   10.1kB
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Deleted: sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3e
Deleted: sha256:a42123db641843401b16bcab8ec0c98e808701d154d1d6c732d99bd445ea8686
Deleted: sha256:4335f5930730b49f9578eb90bf66da5541f838937d85483b27af062a2e926f05
Deleted: sha256:98e01a97a938f2d343a017612a5c5b914f70a6b28f010e0c344f8e14a564d490
Deleted: sha256:9e34eb008884b002306f95430094fd59ea63ddaecb73f7ec20618e9a2daf4b97
Deleted: sha256:45ae75986c980f3218bd6338bde27ea125944ce589ba945d14ce83a0845c354d
Deleted: sha256:8aca92abdb807482cb0ecc82432a0beef58bd4cad0ae94508fe37ee10181cc9f
Deleted: sha256:a257f20c716c7b5d0474b1b5998f4d1090b573d90b82f764b8f1f1faeb4b524f
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
hello-world   latest    1b44b5a3e06a   6 months ago   10.1kB
[ec2-user@ip-172-31-12-159 ~]$
```

```
[ec2-user@ip-172-31-12-159 ~]$ docker inspect nginx
[
    {
        "Id": "sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3e",
        "RepoTags": [
            "nginx:latest"
        ],
        "RepoDigests": [
            "nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e"
        ],
        "Parent": "",
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-02-24T19:05:02.861463578Z",
        "Container": "",
        "ContainerConfig": {
```

```
[ec2-user@ip-172-31-12-159 ~]$ docker search nginx
NAME                                      DESCRIPTION                                     STARS     OFFICIAL
nginx                                     Official build of Nginx.                        21191     [OK]
nginx/nginx-ingress                       NGINX and  NGINX Plus Ingress Controllers fo…   115
nginx/nginx-prometheus-exporter           NGINX Prometheus Exporter for NGINX and NGIN…   50
nginx/unit                                This repository is retired, use the Docker o…   66
nginx/nginx-ingress-operator              NGINX Ingress Operator for NGINX and NGINX P…   3
nginx/nginx-quic-qns                      NGINX QUIC interop                              1
nginx/nginxaas-loadbalancer-kubernetes                                                    1
nginx/unit-preview                        Unit preview features                           0
bitnami/nginx                             Bitnami Secure Image for nginx                  203
bitnamicharts/nginx                       Bitnami Helm chart for NGINX Open Source        3
ubuntu/nginx                              Nginx, a high-performance reverse proxy & we…   139
kasmweb/nginx                             An Nginx image based off nginx:alpine and in…   8
rancher/nginx                                                                             3
linuxserver/nginx                         An Nginx container, brought to you by LinuxS…   236
dtagdevsec/nginx                          T-Pot Nginx                                     0
vmware/nginx                                                                              3
paketobuildpacks/nginx                                                                    0
chainguard/nginx                          Build, ship and run secure software with Cha…   5
gluufederation/nginx                       A customized NGINX image containing a consu…   1
cleanstart/nginx                          Secure by Design, Built for Speed, Hardened …   0
antrea/nginx                              Nginx server used for Antrea e2e testing        0
intel/nginx                                                                               0
circleci/nginx                            This image is for internal use                  2
docksal/nginx                             Nginx service image for Docksal                 1
corpusops/nginx                           https://github.com/corpusops/docker-images/     1
[ec2-user@ip-172-31-12-159 ~]$
```

3. Docker Containers:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker run -d -p 80:80 --name webserver nginx
8faccc0aa87f79bc7f552d0d1f8aae7aa933c148a1d91a202240035df8f234be
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
8faccc0aa87f   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 4 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker stop webserver
webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS                     PORTS     NAMES
8faccc0aa87f   nginx     "/docker-entrypoint.…"   19 seconds ago   Exited (0) 9 seconds ago             webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker start webserver
webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
```




```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
8faccc0aa87f   nginx     "/docker-entrypoint.…"   4 minutes ago   Up 4 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp   webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker stop webserver
webserver
[ec2-user@ip-172-31-12-159 ~]$ docker rm webserver
webserver
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
```

4. **Docker volume:**

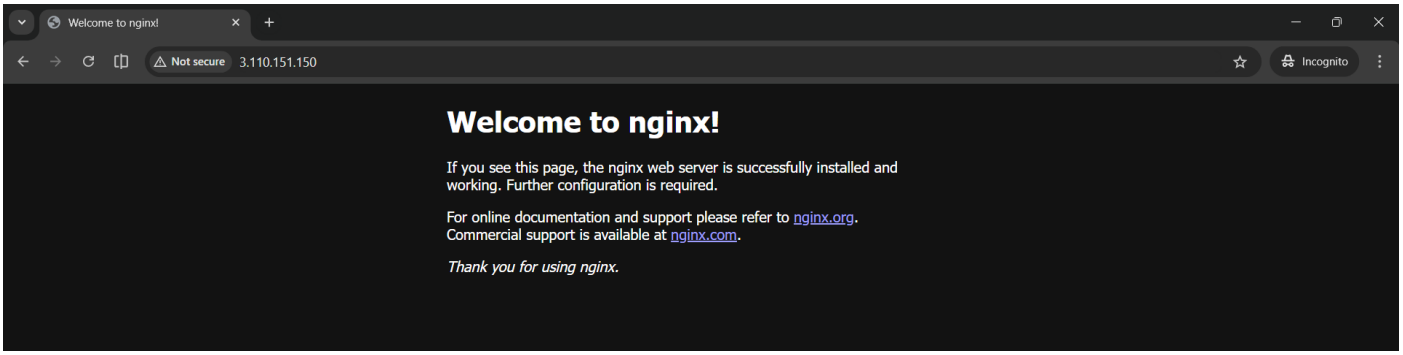
Creating a docker volume named nginx-volume:

```
ec2-user@ip-172-31-12-15!

[ec2-user@ip-172-31-12-159 ~]$ docker volume ls
DRIVER    VOLUME NAME
[ec2-user@ip-172-31-12-159 ~]$ docker volume create nginx-volume
nginx-volume
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker volume ls
DRIVER    VOLUME NAME
local     nginx-volume
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker volume inspect nginx-volume
[
    {
        "CreatedAt": "2026-02-27T11:03:09Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/nginx-volume/_data",
        "Name": "nginx-volume",
        "Options": null,
        "Scope": "local"
    }
]
```

Creating a Container using the volume nginx-volume:

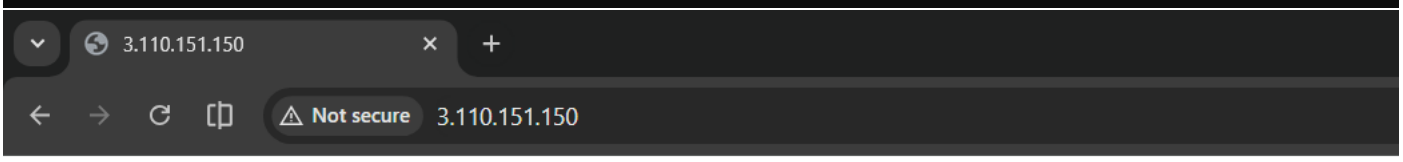
```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker run -d -p 80:80 -v nginx-volume:/usr/share/nginx/html --name mynginx nginx
23a12ffa7c4c5dfb9e7887628d83e2b1126f12eef1eba57aab27d4d0c5a13241
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
23a12ffa7c4c   nginx     "/docker-entrypoint.…"   3 seconds ago   Up 3 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   mynginx
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
```

Verifying the Container status:



Changing the index.html file inside the volume mounted:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker exec mynginx sh -c "echo '<h1>Hello from Within the Volume Nginx-Volume</h1>' > /usr/share/nginx/html/index.html"
[ec2-user@ip-172-31-12-159 ~]$
```



Hello from Within the Volume Nginx-Volume

Now going to delete the container and create a new container with the same volume to check if the files are still the same:

Removing the Container:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ ls
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
23a12ffa7c4c   nginx     "/docker-entrypoint.…"   9 minutes ago   Up 9 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp       mynginx
[ec2-user@ip-172-31-12-159 ~]$ docker stop mynginx
mynginx
[ec2-user@ip-172-31-12-159 ~]$ docker rm mynginx
mynginx
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
```

Creating new Container with the same volume:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker run -d -p 80:80 -v nginx-volume:/usr/share/nginx/html --name new_container_test_volume nginx
de778eec002cc2affba5521275a00e6b2b5b2d96dc0505f2cc76c2d01b09cbcb
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
de778eec002c   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 3 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp       new_container_test_volume
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
```

3.110.151.150

Not secure 3.110.151.150

Hello from Within the Volume Nginx-Volume

Removing the container and checking the data on the volume and removing the volume:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                   NAMES
de778eec002c   nginx     "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:80->80/tcp, :::80->80/tcp       new_container_test_volume
[ec2-user@ip-172-31-12-159 ~]$ docker stop new_container_test_volume

new_container_test_volume
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker rm new_container_test_volume
new_container_test_volume
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ sudo ls /var/lib/docker/volumes/nginx-volume/_data
50x.html  index.html
[ec2-user@ip-172-31-12-159 ~]$ sudo cat /var/lib/docker/volumes/nginx-volume/_data/index.html
<h1>Hello from Within the Volume Nginx-Volume</h1>
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker volume rm nginx-volume
nginx-volume
[ec2-user@ip-172-31-12-159 ~]$ docker volume ls
DRIVER    VOLUME NAME
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
```

5. **Docker Network:**

Creating a new bridge network and creating a container using this newly created network.

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
e249eaec6a4d   bridge    bridge    local
687716f91d8e   host      host      local
625bef45f024   none      null      local
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker network create my-secure-net
5ffb4fe6c83bcabb82366ffcc13f8ad8b06dab8c54fba6b0ea779ab760e447a0
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
e249eaec6a4d   bridge          bridge    local
687716f91d8e   host            host      local
5ffb4fe6c83b   my-secure-net   bridge    local
625bef45f024   none            null      local
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker run -d --name nginx_in_network --network my-secure-net nginx
38e627b7b151d514107d44699d6413f35f45f131feb73982fc21905bd01cac82
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS     NAMES
38e627b7b151   nginx     "/docker-entrypoint.…"   5 seconds ago   Up 3 seconds   80/tcp    nginx_in_network
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker network inspect my-secure-net
[
    {
        "Name": "my-secure-net",
        "Id": "5ffb4fe6c83bcabb82366ffcc13f8ad8b06dab8c54fba6b0ea779ab760e447a0",
        "Created": "2026-02-27T12:17:47.148864268Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.20.0.0/16",
                    "Gateway": "172.20.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "38e627b7b151d514107d44699d6413f35f45f131feb73982fc21905bd01cac82": {
                "Name": "nginx_in_network",
                "EndpointID": "0730532e71164acb8f6c1399731edc6534121784011078ed1b7671df2a8e588c",
                "MacAddress": "02:42:ac:14:00:02",
                "IPv4Address": "172.20.0.2/16",
                "IPv6Address": ""
            }
        },
        "Options": {},
        "Labels": {}
    }
]
[ec2-user@ip-172-31-12-159 ~]$
```

Inspecting the network after disconnecting it from the container and connecting it to the container:

```
ec2-user@ip-172-31-12-159:~

[ec2-user@ip-172-31-12-159 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                       NAMES
55642fc59e9d   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 3 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   nginx_in_network
[ec2-user@ip-172-31-12-159 ~]$ docker network disconnect my-secure-net nginx_in_network
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker network inspect my-secure-net
[
    {
        "Name": "my-secure-net",
        "Id": "5ffb4fe6c83bcabb82366ffcc13f8ad8b06dab8c54fba6b0ea779ab760e447a0",
        "Created": "2026-02-27T12:17:47.148864268Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.20.0.0/16",
                    "Gateway": "172.20.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {}
    }
]
[ec2-user@ip-172-31-12-159 ~]$ docker network connect my-secure-net nginx_in_network
[ec2-user@ip-172-31-12-159 ~]$
[ec2-user@ip-172-31-12-159 ~]$ docker network inspect my-secure-net
[
    {
        "Name": "my-secure-net",
        "Id": "5ffb4fe6c83bcabb82366ffcc13f8ad8b06dab8c54fba6b0ea779ab760e447a0",
        "Created": "2026-02-27T12:17:47.148864268Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.20.0.0/16",
                    "Gateway": "172.20.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "55642fc59e9daee77c6680764a775eb0df59298305a47452f36483f9a54d9aa0": {
                "Name": "nginx_in_network",
                "EndpointID": "5898a71d02a730b409131173e82de2a162547ef097364f2ffa405ccbceb56435",
                "MacAddress": "02:42:ac:14:00:02",
                "IPv4Address": "172.20.0.2/16",
                "IPv6Address": ""
            }
```

*****END*****